THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Marks of 93, 80 and 57 are given weights 3, 2 and 6 respectively. The weighted average is: 1

(a) 70 **(b)** 72 **(c)** 73 **(d)** 71

2. 260 mL of a 33% sugar solution is mixed with 200 mL of a 10% sugar solution. The concentration of the mixture is: 1

(a) 23% **(b)** 25% **(c)** 22% **(d)** 24%

3. In a 100% stacked bar of a household's monthly water use, appliances take 25% of 1140 units. The units used by the appliances are: 1

(a) 285 **(b)** 284 **(c)** 287 **(d)** 286

4. Mixing two solutions of different concentrations gives a concentration that lies: 1

(a) equal to the larger one **(b)** above both

(c) below both **(d)** between the two concentrations

5. The average of 30 numbers is 39. The average of all 42 numbers (including 12 more numbers) is 47. The average of the 12 additional numbers is: 1

(a) 68 **(b)** 66 **(c)** 69 **(d)** 67

6. The average of 15 numbers is 56. If the number 84 is removed, the average of the remaining 14 numbers is: 1

(a) 56 **(b)** 54 **(c)** 53 **(d)** 55

7. The average marks of 58 students is 46 and of another 29 students is 37. The average of all 87 students is: 1

(a) 44 **(b)** 45 **(c)** 42 **(d)** 43

8. The average of 9 numbers is 60. When a number 40 is included, the average of all 10 numbers becomes: 1

(a) 58 **(b)** 60 **(c)** 57 **(d)** 59

9. Marks of 50, 68 and 88 are given weights 6, 3 and 3 respectively. The weighted average is: 1

(a) 63 **(b)** 65 **(c)** 64 **(d)** 66

10. 80 mL of a 10% sugar solution is mixed with 260 mL of a 27% sugar solution. The concentration of the mixture is: 1

(a) 23% **(b)** 25% **(c)** 22% **(d)** 24%

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A final score combines Maths (score 89, weight 2), Science (score 81, weight 5), English (score 61, weight 5). Find the weighted average score. 2

12. A 100% stacked bar of a school's ₹2700 budget shows 28% on books and 11% on sports. Find the amount on books, on sports and on the remaining items. 2

13. The average marks of 36 students in a test is 60 and that of 52 students is 38. Find the average marks of all 88 students together. 2

14. A final score combines Maths (score 58, weight 4), Science (score 67, weight 4), English (score 66, weight 6). Find the weighted average score. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 320 mL of a 32% sugar solution is mixed with 400 mL of a 41% sugar solution. Find the percentage concentration of the mixture. 3

16. The mean of 16 observations is 40. A new observation 57 is added. Find the new mean. 3

17. The average score of 32 students in a test is 47. When 16 more students take the test, the average score of all 48 students becomes 43. Find the average score of the 16 additional students. 3

OR

420 mL of a 18% sugar solution is mixed with 330 mL of a 43% sugar solution. Find the percentage concentration of the mixture. 3

18. The mean of 11 observations is 46. One observation 76 is removed. Find the mean of the remaining observations. 3

OR

The average score of 12 students in a test is 52. When 36 more students take the test, the average score of all 48 students becomes 31. Find the average score of the 36 additional students. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three sections have 30, 35 and 35 students with average marks 73, 52 and 54 respectively. Find the average marks of all the students. 5

OR

A family spent ₹2000 in January and ₹2900 in February. Food took 47% of the January total and 25% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

20. Three sections have 32, 29 and 16 students with average marks 42, 41 and 39 respectively. Find the average marks of all the students. 5

OR

A family spent ₹2000 in January and ₹2100 in February. Food took 27% of the January total and 22% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

END OF QUESTION PAPER — Total: 40 Marks